

Cell Biology - Chapter 2: The Cell

The cell is the basic unit of life. All living organisms are made of one or more cells.

Prokaryotic cells, such as bacteria, lack a nucleus and membrane-bound organelles.

Eukaryotic cells have a nucleus that contains the cell's DNA.

The mitochondrion produces ATP through cellular respiration; it is the powerhouse of the cell.

Ribosomes synthesize proteins by translating messenger RNA.

The endoplasmic reticulum (ER): rough ER is studded with ribosomes, smooth ER makes lipids.

The Golgi apparatus modifies, sorts and packages proteins for secretion.

Lysosomes contain digestive enzymes that break down waste and worn-out organelles.

The cell membrane is a phospholipid bilayer that controls what enters and leaves the cell.

Chloroplasts, found in plant cells, carry out photosynthesis.

The cytoskeleton is a network of protein fibers that gives the cell its shape.

Diffusion is the movement of molecules from high to low concentration.

Osmosis is the diffusion of water across a selectively permeable membrane.

Active transport moves substances against their gradient and requires ATP.